UNIT 4 · TOPIC 5(B)

Gearbox and Travel-Speed Calculations

Syllabus topic: Tractor power transmission: clutch, gearbox, differential and final drives

Gearbox types

Type	Feature
SLIDING-MESH	Oldest; straight spur gears; double-declutch
CONSTANT-MESH	Gears always engaged; DOG CLUTCHES select the ratio
SYNCHROMESH	Cone synchronisers equalise speeds — quiet on-the-move shifts
POWERSHIFT / shuttle	Clutchless ratio or direction change under load
CVT / IVT	Top end

- Indian tractors: commonly **8 forward + 2 reverse**, some 12F + 3R.
- **Creeper gear** (< 1 km/h) for special work — transplanting, cane planting.

Speed and torque at the wheels

$$\text{Wheel rpm} = \text{engine rpm} \div \text{total reduction; travel speed } v = \pi \times D_{wheel} \times N_{wheel}$$

$$\text{Wheel torque} = \text{engine torque} \times \text{reduction} \times \text{transmission efficiency}$$

Example. Engine 2200 rpm, total reduction 60:1 → wheel = 2200/60 = **36.7 rpm**. Wheel D = 1.35 m → v = π × 1.35 × 36.7 = 155.6 m/min ≈ **9.3 km/h**. Wheel torque = 195 N·m × 60 × 0.9 ≈ **10.5 kN·m**.

Practice MCQs

Q115. Gear speeds are matched by friction cones in a:

(a) Sliding-mesh box (b) Synchromesh gearbox
(c) Chain drive (d) Belt CVT

Q116. In a constant-mesh gearbox, ratios are selected by:

(a) Sliding spur gears (b) Dog clutches
(c) Belts (d) Hydraulic rams

Q117. A creeper gear provides speeds of about:

(a) <1 km/h (b) 5 km/h
(c) 15 km/h (d) 30 km/h

Q118. Engine 2200 rpm, total reduction 60:1 → wheel speed ≈

(a) 18.3 (b) 36.7
(c) 55 (d) 73 rpm

Q119. With 1.35-m wheels at 36.7 rpm, travel speed ≈

(a) 4.7 (b) 9.3
(c) 14 (d) 18.6 km/h

Q120. Wheel torque = engine torque ×

(a) Reduction only
(b) Reduction × transmission efficiency
(c) Efficiency only
(d) Speed

Answers

Q115 — (b) Synchromesh gearbox
Q116 — (b) Dog clutches
Q117 — (a) <1 km/h
Q118 — (b) 36.7 — *2200/60 = 36.7 rpm.*
Q119 — (b) 9.3 — *v = π × 1.35 × 36.7 = 155.6 m/min = 9.3 km/h.*
Q120 — (b) Reduction × transmission efficiency

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; IS 4931/ISO 500 (PTO), IS 4468/ISO 730 (linkage); NPTEL farm power lectures.